GMC LANKA CREATIONS

Test Coverage Report – Sprint 8

SENG 34213 – System Development Project

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Version history

Version No.	Date	Author	Summary of Changes
01	2026/10/24	Team	Coverage after Sprint 8 (release 0.1.1)

CHAPTER 1 COVERAGE SUMMARY

This report summarises code coverage at the end of Sprint 8, measured on 24 October 2026. Backend coverage is measured by JaCoCo 0.8.15; the CI pipeline merges the data from unit tests (Surefire) and API integration tests (Failsafe) and fails the build if merged line coverage is below 80%. The model package (JPA entities) and BackendApplication are excluded. Frontend coverage is measured by Vitest with the V8 provider.

Table 1.1: Backend coverage

Measure	**Unit tests only**	**Unit + integration (merged)**	**Gate / target**	**Result**
Line coverage	70.3%	91.3% (1819/1993)	≥ 80% (CI gate); ≥ 70% (NFR014)	Met
Branch coverage	58.9%	74.3% (486/654)	90% for critical business logic	Not met
Instruction coverage	67.0%	90.2%	—	—
Method coverage	55.4%	89.7%	—	—

Table 1.2: Coverage by package (%)

Package	**Lines (unit)**	**Lines (merged)**	**Branches (merged)**	**Methods (merged)**
config	81.9	86.2	12.5	84.0
controller	2.5	100.0	78.0	100.0
exception	73.2	92.9	n/a	80.0
security	44.6	86.1	60.0	90.0
service	93.8	100.0	100.0	100.0
service.impl	78.2	90.8	76.9	87.7

CHAPTER 2 COVERAGE BY CLASS

Table 2.1: Merged coverage by class

Package	Class	Lines	Line %	Branch %
config	AsyncConfig	1	100.0	n/a
config	OpenApiConfig	20	60.0	0.0
config	PasswordConfig	2	100.0	n/a
config	SchedulingConfig	8	62.5	50.0
config	SchemaFixRunner	6	66.7	n/a
config	SecurityConfig	53	100.0	n/a
config	WebConfig	4	100.0	50.0
controller	AdminController	61	100.0	79.2
controller	AppointmentController	8	100.0	100.0
controller	AuthController	26	100.0	70.0
controller	CartController	6	100.0	n/a
controller	ComplaintController	2	100.0	n/a
controller	ContactController	7	100.0	n/a
controller	CustomDesignController	3	100.0	n/a
controller	FeedbackController	3	100.0	n/a
controller	OrderController	6	100.0	n/a
controller	PaymentController	10	100.0	75.0
controller	ProductController	16	100.0	80.0
controller	TechnicianController	2	100.0	n/a
controller	UserController	13	100.0	n/a
exception	BusinessRuleException	2	100.0	n/a
exception	DuplicateEmailException	2	100.0	n/a
exception	GlobalExceptionHandler	94	91.5	n/a
exception	InvalidFileException	2	100.0	n/a
exception	InvalidOrderStatusTransitionException	2	100.0	n/a
exception	InvalidTokenException	4	100.0	n/a
exception	ResourceNotFoundException	2	100.0	n/a
exception	SecurityBreachException	2	100.0	n/a
exception	TooManyRequestsException	2	100.0	n/a
security	CustomUserDetails	14	85.7	100.0
security	CustomUserDetailsService	4	100.0	n/a
security	JwtAuthenticationFilter	26	92.3	59.1
security	JwtTokenProvider	40	85.0	64.3
security	RateLimiter	17	76.5	50.0
service	CheckoutPricing	11	100.0	100.0
service	CheckoutPricing.PriceBreakdown	1	100.0	n/a
service	RefreshTokenResult	1	100.0	n/a
service	StripeGateway.Intent	3	100.0	n/a
service.impl	AppointmentServiceImpl	165	98.8	86.2
service.impl	AuthServiceImpl	63	100.0	75.0
service.impl	CartServiceImpl	82	97.6	81.2
service.impl	CategoryServiceImpl	25	92.0	75.0
service.impl	ComplaintServiceImpl	76	97.4	78.6
service.impl	ContactServiceImpl	10	100.0	75.0
service.impl	CustomDesignServiceImpl	55	89.1	66.7
service.impl	EmailServiceImpl	20	20.0	0.0
service.impl	FeedbackServiceImpl	105	97.1	88.1
service.impl	FileStorageServiceImpl	24	87.5	72.2
service.impl	InvoiceServiceImpl	32	78.1	66.7
service.impl	MaintenanceServiceImpl	96	99.0	90.0
service.impl	NotificationServiceImpl	47	95.7	91.7
service.impl	OrderServiceImpl	363	97.2	82.1
service.impl	PasswordResetServiceImpl	36	94.4	100.0

Package	Class	Lines	Line %	Branch %
service.impl	ProductServiceImpl	101	99.0	86.8
service.impl	RefreshTokenServiceImpl	51	88.2	66.7
service.impl	StripeGatewayImpl	81	8.6	0.0
service.impl	UserServiceImpl	75	100.0	83.3

CHAPTER 3 FRONTEND COVERAGE

Per-file thresholds of 80% (lines, branches, functions and statements) are enforced on the frontend files that contain shared logic. All of them pass (Table 3.1).

Table 3.1: Gated frontend files

File	Lines %	Branches %	Functions %	Statements %
api/axios.js	97.67	83.33	100	97.82
components/common/Pagination.jsx	94.11	96.96	92.85	96.15
utils/deadline.js	100	100	100	100
utils/formatters.js	100	100	100	100

Overall frontend coverage is 4.92% of lines (96/1951), 4.47% of branches and 4.6% of functions, because page components are not covered by automated tests. Component tests for the main pages are planned as future work.

CHAPTER 4 OBSERVATIONS AND ACTIONS

- Backend line coverage (91.3%) exceeds the CI gate; integration tests add 21.0 percentage points over unit tests alone, mainly in controllers and security classes.
- Branch coverage of the service layer is below the 90% target for critical logic. Lowest classes: CustomDesignServiceImpl (66.7%), InvoiceServiceImpl (66.7%), RefreshTokenServiceImpl (66.7%), FileStorageServiceImpl (72.2%), ContactServiceImpl (75.0%).
- EmailServiceImpl and StripeGatewayImpl show 0% branch coverage because they are adapters to external services that are mocked in all tests.
- The config package has low branch coverage because its branches handle start-up conditions that occur only in production (missing keys, schema repair).
- Action: add unit tests for error branches of the payment and appointment services, and introduce component tests for the checkout, cart and order pages.

The HTML coverage reports are produced by CI as build artifacts (coverage-report) and uploaded to Codecov.